

OMER GADALLA

My Portfolio

Galaxy Rocket Team

My passion for engineering began at a young age and inspired me to pursue a degree in Aerospace Engineering. In 2024, I joined the Galaxy Rocket Team, working towards competing in the TEKNOFEST Rocket Competitions. Through this experience, I gained hands-on expertise in **rocket design, Bipropellant rocket engine design and system engineering**. Balancing with METU's hard coursework strengthened both my technical foundation and my ability to perform under pressure. Here are some of my projects.



Project 1: TEKNOFEST Liquid Bipropellant Rocket Engine Development Competition (B3) (2025–2026)

With my team we joined TEKNOFEST Liquid Bipropellant Rocket Engine competition to design and test a fully functional LOX/Jet-A1. The engine needed to have 100N of thrust and combustion at 3000 Kelvin. The competition's duration was 2 years, first year was Research & Development. Second year was subsystem tests and engine ignition.

My Actions in the project

I was the leader of the analysis team. Every design that was designed during the process was passed to me and I performed its analysis. I used to perform the analysis then discuss it with the system engineer and members from the design team to optimize the system as much as we can without increasing the budget madly. I first compared this analysis with the theoretical calculations, then I compared it with the cold flow tests, at the end I compared the analysis with the hot fire test.

The components that were analyzed:

Combustion Chamber

- CFD Combustion Analysis
- CFD Flow Analysis
- Thermo-Structural Analysis



Figure 2 Combustion Chamber



Figure 1 Co-axial Swirl Injector

Co-axial Swirl Injector

- CFD Multiphase Volume of Fluid (VOF) Analysis
- CFD Discrete Phase Method (DPM) Analysis

I gained experience in some of the most critical fields in Aerospace Engineering Industry like:

- Combustion Technology
- Droplet Dynamics
- High Temperature High Pressure Systems
- Swirl Injectors
- Combustion Chamber design, analysis and test

Some of the analysis I conducted:

FLOW Analysis

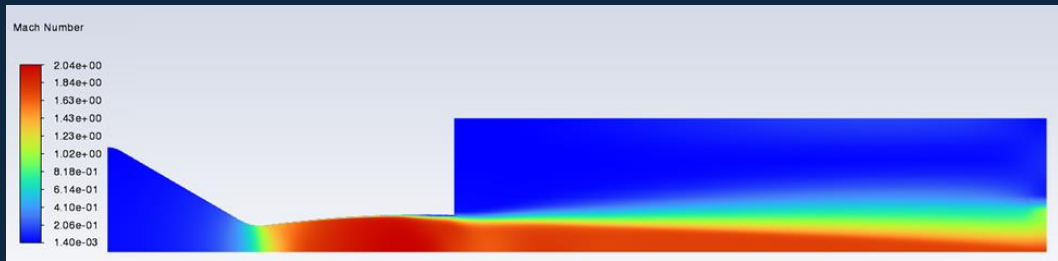


Figure 4 Combustion Chamber and Nozzle Flow Analysis

FUEL Atomizer Multiphase (VOF) Analysis

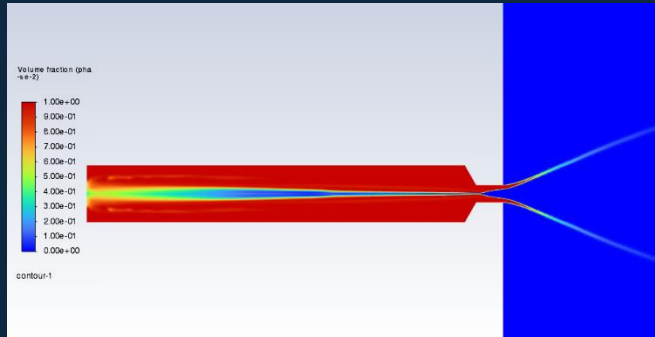


Figure 5 Fuel VOF

OXIDIZER Atomizer Multiphase (VOF) Analysis

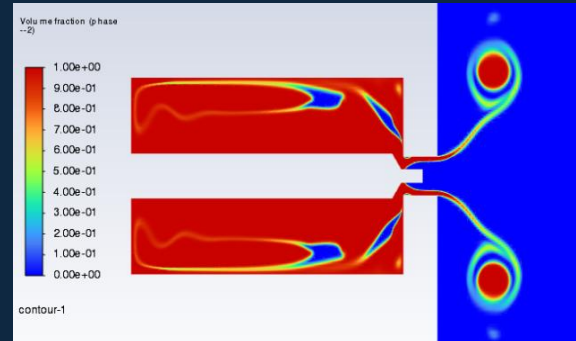


Figure 6 Oxidizer VOF

COMBUSTION Analysis

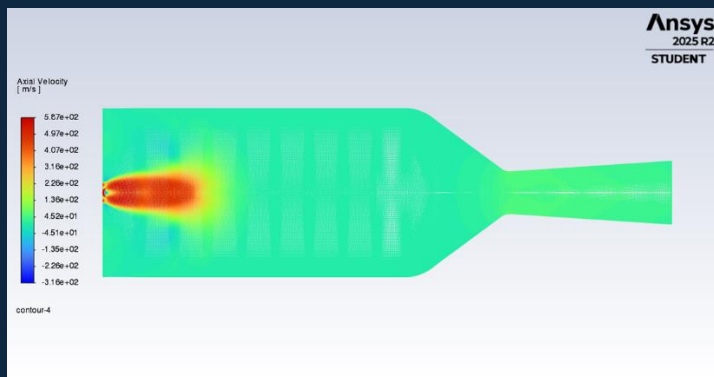


Figure 7 Combustion Analysis

INJECTOR DPM Analysis

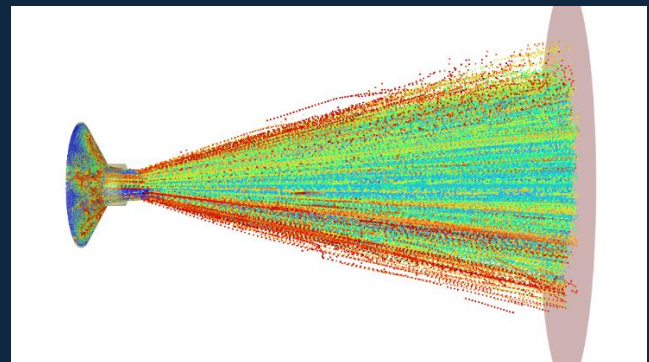


Figure 8 Atomization Analysis

TRANSIENT THERMAL Analysis

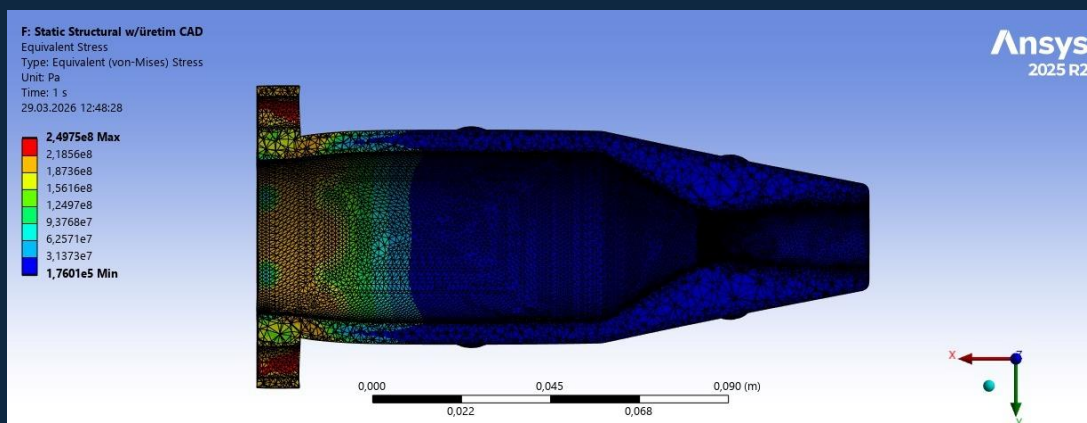


Figure 9 Transient Thermal Analysis

Project 2: TEKNOFEST High Altitude Rocket Competition (2025)

For the TEKNOFEST rocket competition, our team was tasked with designing a high-altitude rocket capable of reaching 15,000 ft, deploying a parachute, and ensuring the rest of the rocket landed safely without any damage to its subsystems.

Our Task: We designed and built a 2-meter rocket capable of reaching Mach 1.4, soaring to 15,000 ft and landing safely.

My Actions in the project:

Position: CFD & Structural Analysis Team Lead

- Lead **aerodynamic** and **structural** design, ensuring performance and safety under high loads.
- Conducted **CFD** simulations in ANSYS Fluent to calculate drag at peak conditions.
- Imported results into ANSYS Mechanical for **structural** and **buckling** analysis.
- Selected **Aluminum 7075** for motor block based on load requirements.
- Integrated composites (carbon fiber, glass fiber, aluminum) to balance weight and strength.

Result: Produced an **optimized rocket structure** with validated aerodynamic and structural safety margins.



Figure 11 The High-Altitude Rocket



Figure 10 Rocket CFD

Gained Experience

- Hands-on expertise in rocket design projects.
- External aerodynamic **CFD analysis**.
- **Composite structure** design and analysis.
- Structural analysis.

Project 3: Sustainable 20N Bipropellant Thruster

During my internship at MTS Defense, the company aimed to enter Türkiye's growing space industry. So I decided to do the conceptual design of a cost-effective, green-propellant thruster for small satellites (10–100 kg) to showcase feasibility and innovation.

My Actions:

R&D Engineer

- Arranged a roadmap to manage development and follow progress.
- Conducted R&D on propulsion subsystems and mission requirements.
- Performed trade-off analysis then selecting N_2O /Propylene for safety and sustainability.
- Designed CAD model, P&ID, and set chamber pressure (10 bar) and thrust target (20 N).

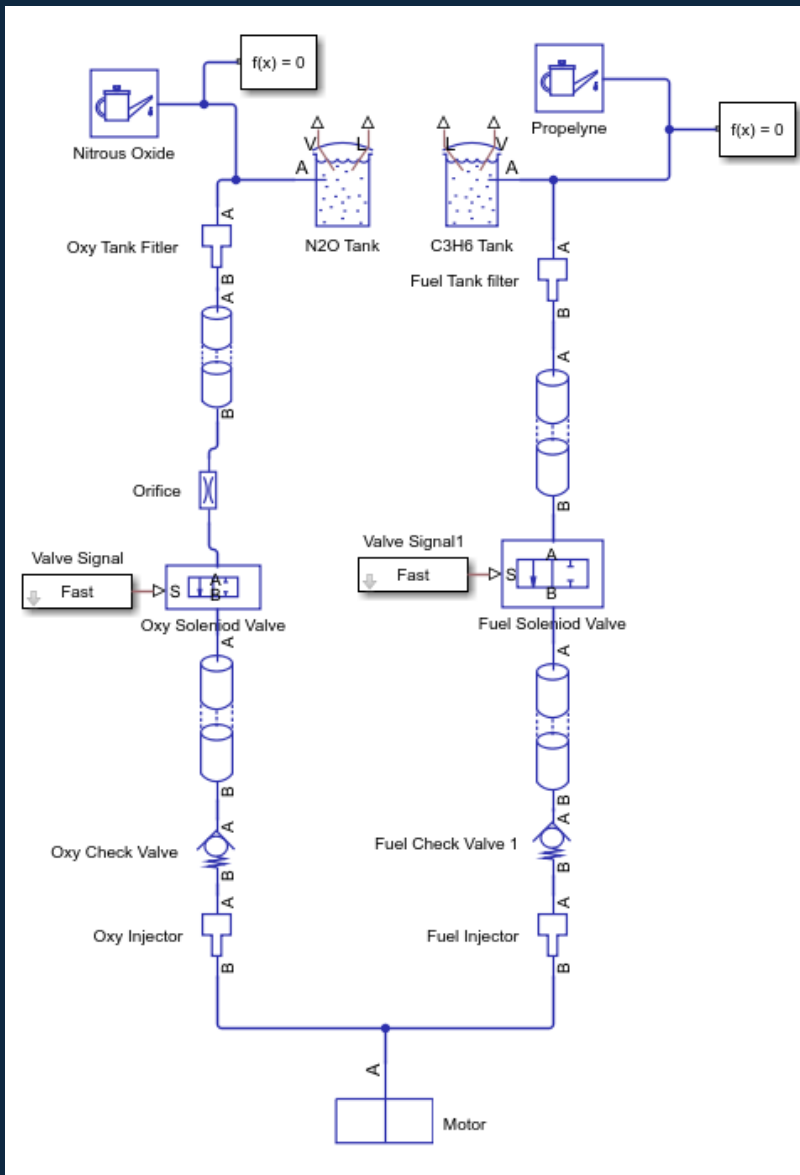


Figure 12 Thruster Simscape Model

Result: Delivered a complete conceptual design package that demonstrated the company's potential entry into the smallsat propulsion market and aligned with sustainable space initiatives.

Gained Experience:

- 1) Knowledge of sustainable aviation fuels.
- 2) Design and production of propellants and gas systems.
- 3) Familiarity with **spacecraft subsystems**.
- 4) Simulation and Modeling of satellite propulsion subsystems in Simscape.
- 5) Familiarity with instrumentation and control components.



Figure 13 3D-Printed Nozzle prototype in hands

Project 4: CFD Analysis of a Converging-Diverging Nozzle (ASE 342 – Aerodynamics II)

Situation: As part of Aerodynamics II coursework, a CFD analysis of a nozzle flow was assigned.

Task: Validate CFD models against experimental data while reducing computational cost.

Action:

- Choose the nozzle geometry based on NASA's published experiment.
- Built viscous compressible flow simulations in **ANSYS Fluent** with **boundary-layer** capturing.
- Implemented axisymmetric simplification to cut computational time while preserving accuracy.

Result: Achieved **CFD** predictions within 5–10% deviation from NASA data and quantified thrust across flow regimes, proving model reliability.

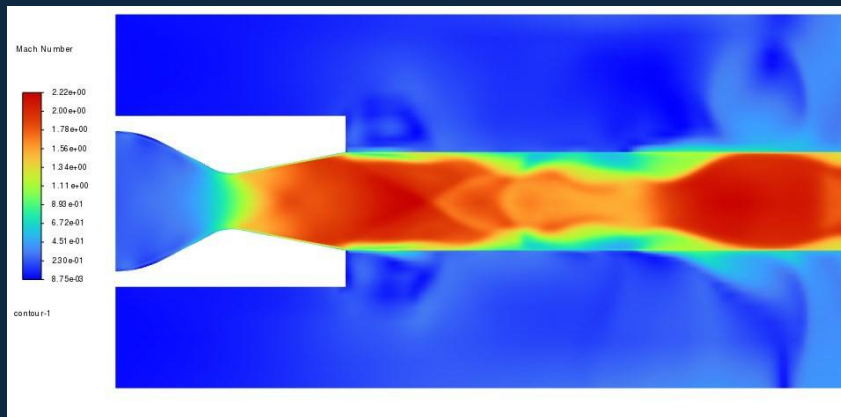


Figure 14 Nozzle CFD Results

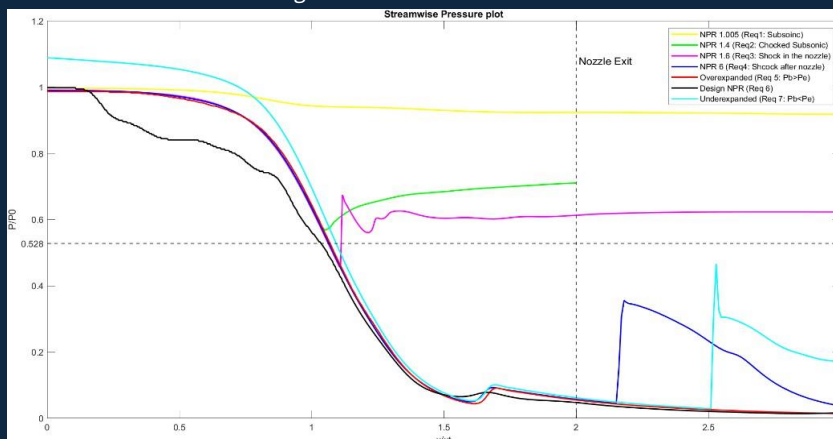


Figure 16 Pressure Distribution Across the Nozzle

Gained Experience

- Acquired hands-on expertise in **CFD simulations**.
- Gained in-depth knowledge of **internal viscous compressible flows** under **high-speed** and **high-pressure** conditions.
- Strengthened understanding of **real-world aerodynamics** through data validation with NASA's experiment.
- Enhanced **technical report writing** and scientific communication skills by preparing detailed documentation of findings.

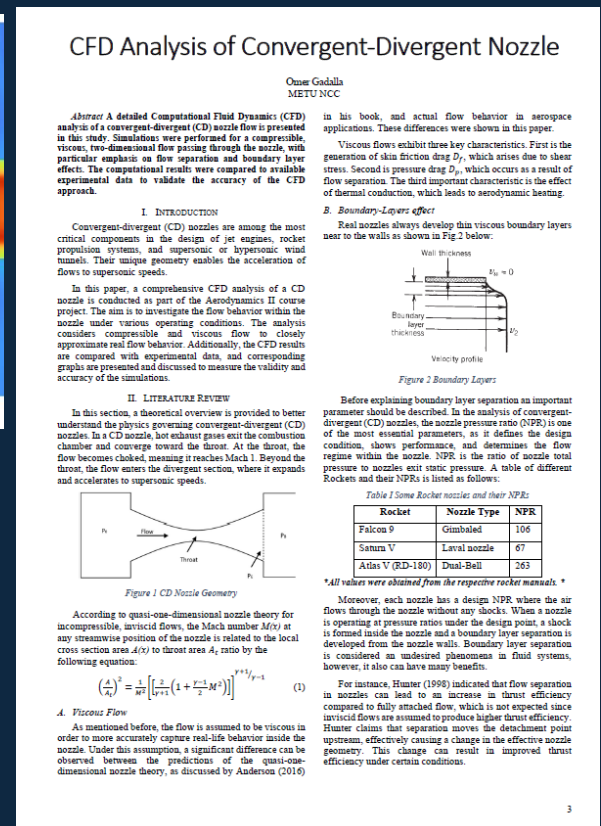


Figure 15 My Academic Paper

Project 5: Rocket Nozzle Design (ASE 435 – Propulsion Systems II)

Situation: Tasked with designing a supersonic nozzle using the Method of Characteristics (MOC).

Task: Develop a MATLAB tool to generate an optimized nozzle contour for Mach 3 exit conditions

Action:

- Implemented MOC for expansion flow fields
- Created Prandtl-Meyer function solver for precise Mach number calculation.

Result: Produced an **optimized nozzle contour** with smooth supersonic expansion, demonstrating practical application of MOC in propulsion design.

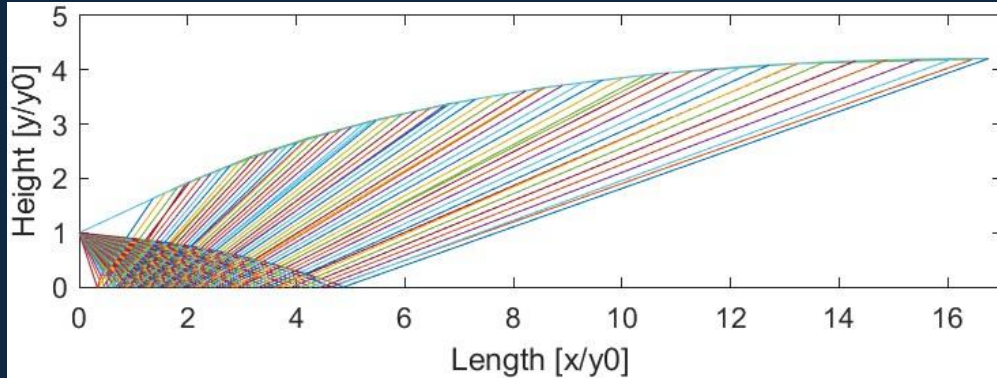


Figure 17 Rocket Nozzle Contour

Project 6: Combustion Analysis of Rocket Combustion (ASE 492 - Rocket Technology)

Situation: The course required simulating equilibrium combustion for a hydrogen-oxygen engine.

Task: Develop a MATLAB-based solver to calculate equilibrium mole fractions of combustion species.

Action:

- Implemented **Newton-Raphson** method and MATLAB's **fsolve** for chemical equilibrium equations.
- Modeled thermodynamic properties for six combustion species.

Results: Created a validated tool that deepened understanding of reaction kinetics and thermodynamics in rocket engines.

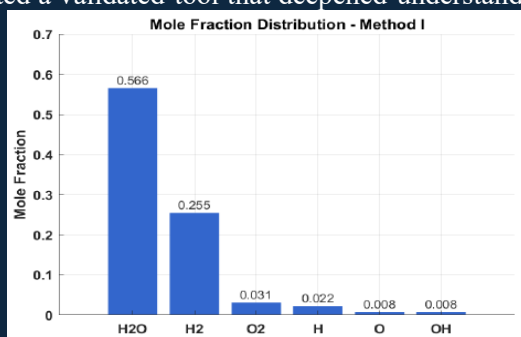


Figure 19 Analysis Results – method I

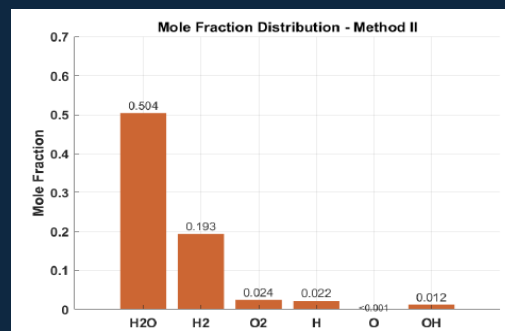


Figure 18 Analysis Results – method II

Gained Experience

- **MATLAB** code writing.
- Knowledge of the **Method of Characteristics**.
- Knowledge of rocket **nozzle optimization**.
- Practical knowledge of **supersonic flow** and **expansion waves**.
- Real-life rocket **combustion analysis**.
- Knowledge of **Numerical Methods**.

My Certifications

MATHWORKS. MATLAB Onramp



MATHWORKS. Simscape Onramp



Rolls-Royce. Reliability and Safety Workshop



Figes A.Ş. MATLAB & Simulink



Udemy. Aerospace Structures and Materials



Udemy. Rocket Science and Engineering

